Saans Saathi

Your Local Air Quality Action Assistant

Understand the **air**. Plan your **day**.

Delhi residents can access air-quality information. But how can they use it to make better-informed daily decisions?

Saans Saathi aims to turn available air-quality forecasts into simple, activity-based guidance.

A proposed platform. Built for Delhi.



THE PROBLEM

Data exists.
Action is missing.

Delhi faces severe winter air pollution episodes impacting daily life.

Measurements, forecasts and advisories exist but are hard to turn into daily decisions.

Air quality varies by location and time, making it complex to interpret.

Students, commuters, outdoor workers, parents and vulnerable groups feel the impact the most.



The gap isn’t data. It’s clear, timely, and actionable guidance for everyday life.

WHY THIS MATTERS AND OUR PROPOSED SOLUTION

From pollution data to everyday decisions.

Saans Saathi turns complex air-quality information into clear guidance you can act on.

01 → 02 → 03 → 04

View available AQI and pollutant measurements for a selected monitoring

Present short-term forecasts.

Select an activity, such as commuting or anyother outdoor exercise, and see relavent advice linked to public health guidance

See a simple explaination of the forecast, its source and its relavant uncertainity where available



EXAMPLE SCENARIO

A student plans an evening run. Saans Saathi displays the available forecast and relevant official guidance, helping the student make a better-informed decision about outdoor exercise.

HOW THE SYSTEM WOULD WORK

From data to guidance, built for daily decisions.

Air-quality data + Weather data → Data cleaning & analysis → Forecasting model → Expected air-quality trend → Guidance from verified advisorys → Simple, user-friendly interface

Forecasting estimates future pollutant levels using historical and current data.

Data analysis identifies recurring local pollution patterns.

Recommendation layer maps forecasted conditions to official health guidance.

Future: Language model may explain information in simple Hindi/English but will not generate independent health advice.



THE AI/ML BEHIND THE IDEA

OUR APPROACH

POLLUTION FORECASTING

Forecast future pollutant levels using regression and time-series models.

AQI CATEGORISATION

Convert forecasts into AQI categories to indicate severity.

RECURRING PATTERNS

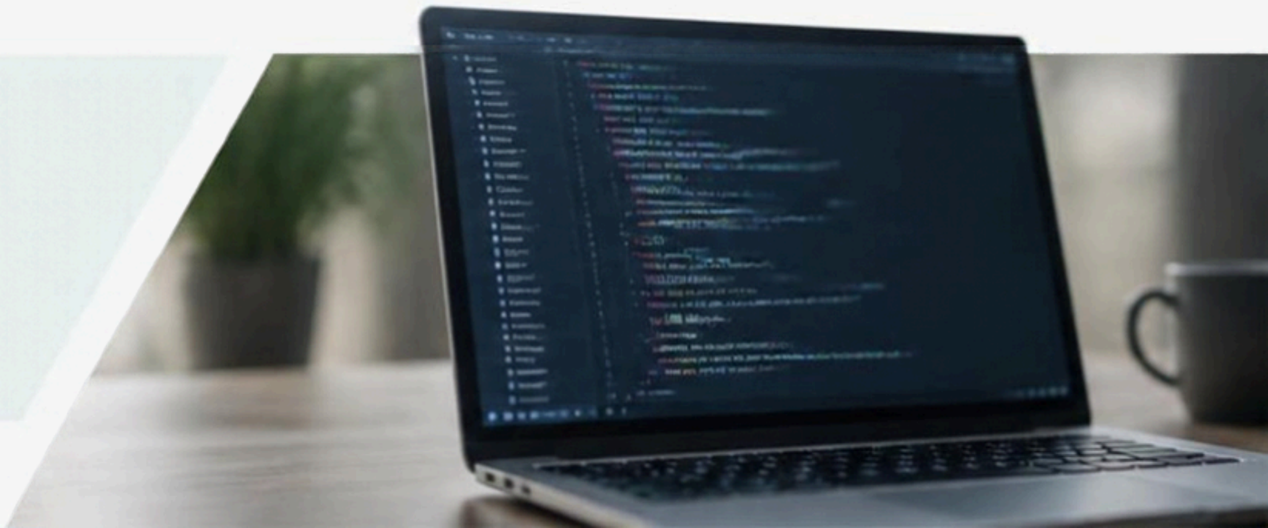
Use clustering and exploratory analysis to find recurring patterns across locations and times.

ACTION EXPLANATIONS

Translate forecasts into clear, practical guidance using a recommendation layer and NLP/LLM.

Start with a baseline forecasting model, evaluate it against actual observations, and add more advanced techniques only if they improve the results.

No reinforcement learning or computer vision needed initially.



Data, Feasibility and Prototype Plan

DATA WE RELY ON

Historical pollutant and AQI records

Monitoring station locations and timestamps

Weather data (wind speed/direction, temperature, humidity, rainfall)

Official AQI categories and public-health advisorys

We start with the CPCB data portal:
https://airquality.cpcb.gov.in/AQI/

and verify availability, licensing, completeness, and API access.

PROTOTYPE ROADMAP

PHASE 1 — Data exploration & basic interface with current/historical data

↓

PHASE 2 — Build and evaluate baseline forecast model

↓

PHASE 3 — Add guidance, alerts, and Hindi/English explanations

Final product would be a web dashboard.



EXISTING SOLUTIONS AND OUR USP

From numbers to meaningful daily choices.

We bridge the gap between official data and real-life decisions.

CPCB AQI portal / SAMEER	Official air-quality info and data	https://airquality.cpcb.gov.in/AQI/
SAFAR	Air-quality forecasting and advisories	http://safari.tropmet.res.in/
Saans Saathi (proposed)	Translate forecasts into clear, activity-focused explanations compare nearby stations clarify uncertainties and sources	Proposed concept only



Expected Impact and Challenges

Saans Saathi helps Delhi residents anticipate air-quality changes and make informed choices — it improves interpretation and use of air-quality information, not pollution levels.

What Saans Saathi can do

Helps residents anticipate air-quality changes

Supports informed choices for outdoor activities

Makes complex data easier to understand

Provides an educational AI/ML prototype for measurable impact

Key Challenges

Missing or inconsistent data

Forecast errors due to atmospheric complexity

Limited spatial coverage between stations

Risk of overconfidence or misinterpretation

Privacy concerns with location or preferences

Existing services may overlap some functionality



RESEARCH AND REFERENCEs

Evidence today. Better decisions tomorrow.

Central Pollution Control Board — National Air Quality Index
https://airquality.cpcb.gov.in/AQI/

SAFAR — System of Air Quality Forecasting and Research
http://safar.tropmet.res.in/

Scientific Reports (2024) — Temporal trends and predictive modeling of air pollutants in Delhi: a comparative study of artificial intelligence models
https://www.nature.com/articles/s.-z

Ministry of Environment, Forest and Climate Change / Press Information Bureau — Air pollution control measures and the National Clean Air Programme
https://www.pib.gov.in/PressReleasePage.aspx?PRID=1989207

ChatGPT for suggested newsletters and references
https://chatgpt.com

NEXT STEP **Verify data availability, compare existing services, and test whether a simple forecasting prototype can provide useful and reliable information.**

